

Instructor: Date:

Learner: Regional Branch:

Program: Center:

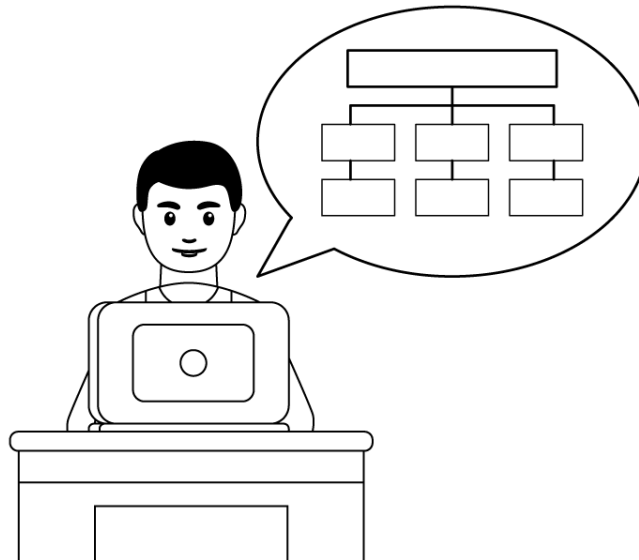
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Let's Start!

Organizing Your Ideas



Learning Outcome:

Exchange information about relational database management system.

Ask your partner:

- How do you organize your ideas when working or studying?
- Do you find mind maps useful to organize information? Explain yourself with an example.

Can Do Statements:

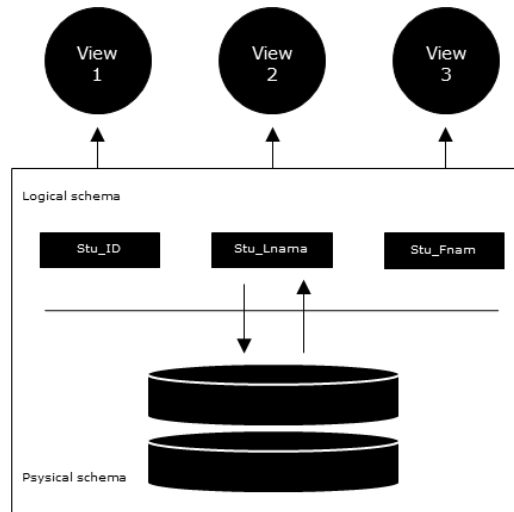
- I can say what a database schema is.
- I can talk about categories in a database schema.
- I can talk about categories in a database schema.





Let's Explore!

Let's clarify some aspects about data schemas in DBMS.



Instructions: Read the text below.

Monica: Dad, I was wondering if you could explain to me some aspects about the data schemas in DBMS.

Frank: Sure. You know I'm an expert on those topics.

Monica: Thank you. It was really difficult for me to keep up with the lessons this week.

Frank: No problem. Let's start with the basics. A database schema defines the logical view of the database; in other words, it is basically the skeleton of the database. Apart from defining its structure, it describes the type of relations among the pieces of information the system holds. It is important for you to know that we develop these schemas before the system, or the data, even exist.

Monica: Great!

Frank: Now, I'd like to know what you learned about the categories of data schemas.

Monica: I think the teacher talked about two different categories.

Frank: That's right. The first category, the physical database schema, relates to the actual storage capacity of the system as well as the way the data is stored. On the other hand, the logical database schema refers to the restraints that the system needs to implement on the system's data.

Monica: Wow! That was pretty clear. However, there is something I still don't understand: Database instance.

Frank: I was wondering if you would ask about that. Well, a database instance basically refers to the status, retrieved at any time, of the operational data of a database. Once the database is fully operational, we can check if the database is in "valid state"; in other words, check if the designers included the needed restraints and conditions that the database initially required.



Let's Practice!

What did Monica explain to Vivian?

Instructions: Proofread the sentences. Correct the mistakes.

1. You have to remember that a database scam defines the logical view of the database.

2. In simple words, it is basically the skull of the whole system.

3. The schema also defines the type of users existent among the pieces of information contained in the system.

4. Last year, I learned that we had to develop the users before the system or the data were created.

5. Before we starting to develop the database, we had to define its structure first.



Let's Practice!

Let's go over* some vocabulary.

*To "go over" something means to review.

Instructions: Find the words in the wordsearch below.

1. The ph_s_c_l schema stores data using files.
2. The physical database schema, relates to the actual s_r_e capacity of the system
3. The _o_i_al schema defines the limitations of data.
4. On the other hand, the logical database schema refers to a set of logical res_ _ _ _ _ .
5. Programmers have to i _ pl _ _ _ _ a set of restraints into the database.

P	J	V	S	J	N	K	D	V	Q	X	H	X	L	W
H	E	L	Y	L	B	O	S	Z	H	C	L	K	L	V
Y	F	J	B	T	R	H	Q	P	W	E	H	T	O	P
S	B	B	T	R	R	E	U	I	U	R	P	R	G	I
I	R	Z	U	O	O	R	D	F	I	G	R	N	I	L
C	Y	R	K	G	B	E	D	I	B	O	P	G	C	G
A	Q	Z	Z	M	U	X	M	M	A	B	T	R	A	L
L	E	X	Y	J	X	Z	Q	E	D	R	M	D	L	Q
K	X	S	E	U	I	O	I	C	J	I	O	C	N	Y
I	M	P	L	E	M	E	N	T	S	C	R	J	H	K
Q	K	L	L	A	W	F	M	O	C	M	H	U	M	U
H	R	E	S	T	R	A	I	N	T	S	Q	P	L	R
W	A	N	A	G	P	E	Z	H	Z	O	U	R	X	N
H	F	Q	Y	I	V	C	J	F	V	G	S	W	E	A
S	T	P	R	A	G	E	I	T	Y	S	K	G	J	U



Extension Activity

Now, let's get to work!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Write the main ideas you learnt about database schemas in a draft paper.
- Propose some questions within the statements you read.
- Design a leaflet which includes all the information above.
- Make the leaflet easy for you to understand.
- Present your leaflet to your partners and instructor.

Self-Assessment

Yes

No

- I identify main ideas about database schemas. ☐ ☐
- I propose meaningful questions within the sentences I read. ☐ ☐
- I create an innovative leaflet that represents the information about database schemas. ☐ ☐
- I present the leaflet to my class and instructor. ☐ ☐



Target Words

- **restraint** (noun): the limitation of information or access within a system
e.g. On the other hand, the logical database schema refers to the restraints that the system needs to implement on the system's data.
- **instance** (noun): something shown as a proof of something
e.g. Well, a database instance basically refers to the status, retrieved at any time, of the operational data of a database.
- **operational** (adjective): something that fully functions
e.g. Once the database is fully operational, we can check if the database is in "valid state"; in other words, check if the designers included the needed restraints and conditions that the database initially required.
- **keep up** (phrasal verbs): to continue doing something
e.g. It was really difficult for me to keep up with the lessons this week.
- **physical** (adjective): related to something that is real or material
e.g. The first category, the physical database schema, relates to the actual storage capacity of the system as well as the way the data is stored.
- **actual** (adjective): current, that is true at the moment of speaking
e.g. The first category, the physical database schema, relates to the actual storage capacity of the system as well as the way the data is stored.



Learning Points

Questions Within Statements

A **question within a statement** is a question that appears in a declarative statement.

You use a question within a statement when you don't want to be direct, like when you are talking to your teacher or someone of higher position; in fact, making a direct question might be impolite.

To make a question within a statement, you need to follow some rules:

1. Introduce the sentence using one of the following expressions: **do you know**, **could you tell me**, and **I wonder**, among others.
2. Use a WH question or "if".
3. Eliminate the auxiliary verb.
4. Set the subject.
5. Use a main verb (conjugated according to the tense) and a complement.
6. Add the question mark or period.

For example:

I was wondering if a schema is necessary for our database.

I don't know why he decided not to help us with the schema.

Examples

- *Dad, I was wondering if you could explain to me some aspects about the data schemas in DBMS.*
- *Now, I'd like to know what you learned about the categories of data schemas.*



Learning Points

Simple Tenses: Review

There are two important tenses that we use in a daily basis: the **present simple** and **past simple**.

PRESENT SIMPLE

Remember that the present simple tense makes reference to different situations:

1. We can use the present simple to talk about routines or habits.
2. We can also use this tense to talk about true facts that never change, such as natural events.
3. We use this tense to give instructions, too.

Remember that when you use this tense, the conjugation of the third person changes as an "-es" or "-s" is required.

For example:

Databases **collect** lots of information.

She **programs** databases every day in her job.

PAST SIMPLE

Now, the past simple tense only makes reference to actions or situations that started and finished in a specific moment in the past. In other words, we deal with situations that are no longer happening.

It is important to remember that the past simple tense requires of special conjugation of its verbs. In fact, depending on the type of verb we have, its conjugation varies. Don't forget that regular verbs just add a final "-ed" or "-d" at the end of the verb to turn it into the past form. On the other hand, irregular verbs do not follow any pattern at all when they change into the past form.

For example:

We **designed** a special database for the company last year.

We **built** the database based on the special needs of the customer.

Examples

- *I think the teacher talked about two different categories.*
- *That's right. The first category, the physical database schema, relates to the actual storage capacity of the system as well as the way the data is stored.*





Learning Points

The Database Schema

Before a database is created, its **schema** has to be outlined and planned.

The process that every database goes through is basically always the same. The first stage in the production of a database is the establishment of the database schema. However, what is a database schema? In simple words, it refers to the structure of the system's logical view. It shapes structure of the database as it establishes its organization as well as the relations created within it.

Database schemas have two different categories:

- **Physical database schema:** Relates to the real storage capacity of the system. In fact, it emphasizes on the way data is stored, which can be either in files or indices. Furthermore, it deals with secondary storage.
- **Logical database schema:** Refers to the strains that the system needs to implement directly on the data that is contained in the system. Last but not least, it deals with any sort of graphs such as tables or views.

Finally, it is important not to forget that database schemas are designed by database designers; once that is done, programmers can proceed with the realization of the database so the final user can use it.

Check the following chart that illustrates this schema:

